

Regular Expression

Course Name: Compiler Design

Course Code: CSE331

Level:3, Term:3

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Regular Expression

A Regular expression (sometimes called a rational expression) is a sequence of characters that define a search pattern, mainly for use in pattern matching with strings, or string matching, i.e. “find and replace”-like operations.

Regular expression is a notation for defining the set of tokens that normally occur in programming languages.

How are patterns specified?

- Define following operators over sets of strings:
 - 1. Union: $L \cup U$
 - $S = L \cup U = \{s \mid (s \in L) \vee (s \in U)\}$
 - 2. Concatenation: LU or $L.U$
 - $S = L . U = \{s t \mid (s \in L) \wedge (t \in U)\}$
 - 3. Kleene closure: L^* , set of all strings of letters, including ϵ ,
 - $S = L^*$ denotes “zero or more concatenations of” L
 - 4. Positive closure: L^+ .
 - $S = L^+$ denotes “one or more concatenation of” L

OPERATION	DEFINITION AND NOTATION
<i>Union of L and M</i>	$L \cup M = \{s \mid s \text{ is in } L \text{ or } s \text{ is in } M\}$
<i>Concatenation of L and M</i>	$LM = \{st \mid s \text{ is in } L \text{ and } t \text{ is in } M\}$
<i>Kleene closure of L</i>	$L^* = \bigcup_{i=0}^{\infty} L^i$
<i>Positive closure of L</i>	$L^+ = \bigcup_{i=1}^{\infty} L^i$

Operations of Language

- Letters or alphabets and digits are the most important elements of language.
- Let L be the set of alphabets $\{A, B, \dots, Z, a, b, \dots, z\}$ and D be the set of digits $\{0, 1, \dots, 9\}$
- L could be in form of upper case and lower case.
- Examples:
 - $L \cup D$ is the set of letters and digits.
 - LD is the set of strings consisting of a letter followed by a digit.
- $LLLL = L^4$ is the set of all four-letter strings.

Operations of Language

- L^* is the set of all strings of letters, including ϵ , the empty string
- $L(L \cup D)^*$ is the set of all strings of letters and digits beginning with a letter.
- D^* is the set of all strings of one or more digits.

Examples

- Let $L = \{a, b\}$
- Some regular expressions:
 - $a \mid b$
 - Denotes the set of $\{a, b\}$ having a or b.
 - $(a \mid b)(a \mid b)$
 - Denotes $\{aa, ab, ba, bb\}$, the set of all strings of a's and b's of length two.
 - a^*
 - Denotes the set of all strings of zero or more a's , i. e., $\{\epsilon, a, aa, aaa, \dots\}$

Examples

- $(a|b)^*$ or $(a^*|b^*)^*$
- Denotes the set of all strings containing zero or more instances of an a or b, that is, the set of all strings of a's and b's.
- $a | a^*b$
- Denotes the set containing the string a and all strings consisting of zero or more a's followed by a b

Language to Regular Expressions

Examples:

- “Set of all strings having at least one ab”
 - $(ab)^+$
- “Set of all strings having even number of aa”
 - $(aaaa)^*$
- “Set of all strings having odd number of bb”
 - $b(bb)^*$
- “Set of all strings having even number of a and even number of b”
 - $(aa)^*(bb)^*$
- “Set of all strings having even number of aa and even number of bb”
 - $(aaaa)^*(bbbb)$

Language to Regular Expressions

- “Set of all strings having zero or more instances of a or b starting with aa”
 - $(aa)(a \mid b)^*$
- “Set of all strings having zero or more instances of a or b ending with bb”
 - $(a \mid b)^* (bb)$
- “Set of all strings having zero or more instances of a or b starting with aa and ending with bb”
 - $(aa) (a \mid b)^* (bb)$

THANK YOU